RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th Dec 2025	**CIE - II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: ENTREPRENEURSHIP & INTELLECTUAL PROPERTY RIGHTS		**Course Code : HS351TA**

Part A

Sl. No	Questions	M	BT	CO
1.	Distinguish between Proof of Concept (PoC) and Prototype with respect to risk mitigation.	2	1	3
2.	Why is technical feasibility meaningless without market feasibility? Justify.	2	1	3
3.	How does reverse engineering weaken trade secret protection?	2	2	3
4.	Why is novelty mandatory but aesthetic appeal critical in industrial design protection?	2	3	3
5.	Define Copyright.	2	2	3

Part B

Sl. No	Questions	M	BT	CO
1.	Explain the elements of a business plan. Describe each component with suitable examples.	10	2	2
2.	Discuss the complete Entrepreneurial Opportunity Evaluation Process covering: • Identification of market opportunities and trends • Technical feasibility (Prototype and Proof of Concept) • Financial feasibility (Cost estimation, revenue projection, and break-even analysis) • Role of cross-disciplinary collaboration in technological innovation	10	3	3
3.	Explain the acts which do not constitute Infringement under the Copyright Law.	10	2	3
4.	Explain in detail the complete procedure for obtaining Industrial Design protection in India. Further, critically evaluate: • Grounds and procedure for revocation of registered designs • Types of design infringement • Civil and criminal remedies available Illustrate your answer with at least one real case study.	10	3	2
5.	A software company develops a unique algorithm for data security and keeps it confidential. A former employee illegally discloses this information to a competitor. 1. Identify the type of intellectual property right involved. 2. Explain the legal remedies available under Indian law. 3. Discuss the importance of trade secret protection with reference to this case.	10	3	3

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**07/01/2025**	**Maximum Marks**	**10 + 50**
Course Code	**HS351TA**	**Duration**	**120 Min**
Sem	**V**	**CIE -II**	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	_____ component is key in assessing market feasibility	1	1	2
2	What does break-even analysis determine?	1	1	3
3	Which strategy in Porter's generic strategies focuses on offering unique products?	1	1	3
4	What does a proof of concept primarily demonstrate?	1	1	2
5	Why is prototyping important in the ideation process?	2	1	2
6	In order to register Industrial Design, Design should be new and _____	1	1	4
7	Can a colour by itself be subject matter of registration of industrial design? If yes, justify your statement.	2	2	4
8	Upon expiry of registration of a design under Designs Act 2000 it becomes _____ property	1	1	4
	Part –B			
1	A medium-sized technology company in India develops an innovative algorithm for optimizing supply chain logistics. To maintain its competitive edge, the company decides not to patent the algorithm but to treat it as a trade secret. Over time, a former employee joins a competitor and allegedly discloses key aspects of the algorithm. The company must navigate the legal landscape to protect its trade secrets. **Questions:** i. What legal tools and mechanisms are available in India to protect trade secrets? ii. Analyze the company's decision to treat the algorithm as a trade secret instead of patenting it. What are the advantages and risks of this approach? What preventive measures could the company have implemented to minimize the risk of trade secret leakage?	10	3	4

2	A furniture design firm creates a unique chair design that combines aesthetics with ergonomic principles. The design becomes a market hit, but a competitor begins manufacturing and selling a chair with an almost identical appearance. The firm initiates legal proceedings to obtain design protection and stop the competitor's activities. Questions: i. What features of the chair design qualify it for industrial design protection?	10	3	4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15/12/2025	**CIE – II-A**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Operations Management		**Course Code : IM352IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	In time series models, regular fluctuations occurring within a year due to weather or festivals are called _____ variations.	1	1	1
2.	The difference between the actual value and the forecasted value is known as the _____.	1	1	2
3.	Qualitative forecasting methods rely heavily on _____ judgment.	1	1	1
4.	A hotel predicts weekend occupancy using past booking patterns. This uses a _____ time series model.	1	2	1
5.	A company uses a senior manager's opinion to forecast sales of a new product. This reflects the _____ forecasting method.	1	1	1
6.	A production line experiencing a delay at one critical machine identifies it as the _____ resource.	1	2	4
7.	A production manager selects the job with the shortest processing time to reduce average waiting time. This applies the _____ rule.	1	2	3
8.	A plant with multiple machines generates several two-machine equivalent sequences to approximate the optimal schedule. This procedure follows the _____.	1	2	3
9.	A manufacturer with two machines creates an optimal job sequence by choosing the smallest processing time among both machines. This applies _____ rule.	1	2	3
10.	For two jobs processed on multiple machines, the job with smaller processing time on the first machine is scheduled _____ in the Johnson-like rule for job shops.	1	2	3
	Part – B			
1	Explain the importance of measuring forecast error in Operations Management. Discuss the managerial implications of each measure with suitable examples.	08	2	3
2	With suitable example, differentiate between forward scheduling and backward scheduling	08	2	4

<table>
<tr><td>3</td><td>A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as 10, 11, 10, 11, 14, 15, 14 & 15 units respectively. What would be his estimate of sales for September?
(i) If he has followed 4 month weighted moving average (with weights 0.4, 0.3, 0.2 & 0.1).
(ii) If he has followed SES with $\alpha = 0.2$ and his estimated sales for January is 11.</td><td>12</td><td>4</td><td>3</td></tr>
<tr><td>4</td><td>Consider the following single machine scheduling problem.
<table>
<tr><th>Job, j</th><th>Processing time, t_j</th><th>Due date, d_j</th></tr>
<tr><td>1</td><td>12</td><td>17</td></tr>
<tr><td>2</td><td>10</td><td>14</td></tr>
<tr><td>3</td><td>10</td><td>12</td></tr>
<tr><td>4</td><td>9</td><td>20</td></tr>
<tr><td>5</td><td>14</td><td>18</td></tr>
<tr><td>6</td><td>17</td><td>27</td></tr>
</table>
Determine the sequence which will minimize the maximum lateness (L_{max}). Also, determine the L_{max} with respect to the optimal sequence.</td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>5</td><td>Use graphical method to minimize the time needed to process the following jobs on the machines shown (i.e. for each machine, find the job which should be scheduled first). Also, calculate the total time elapsed to complete both jobs.
<table>
<tr><td rowspan="2">Job 1</td><td>Sequence</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr>
<tr><td>Time (Hrs).</td><td>3</td><td>4</td><td>2</td><td>6</td><td>2</td></tr>
<tr><td rowspan="2">Job 2</td><td>Sequence</td><td>B</td><td>C</td><td>A</td><td>D</td><td>E</td></tr>
<tr><td>Time (Hrs)</td><td>5</td><td>4</td><td>3</td><td>2</td><td>6</td></tr>
</table></td><td>12</td><td>4</td><td>4</td></tr>
</table>

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

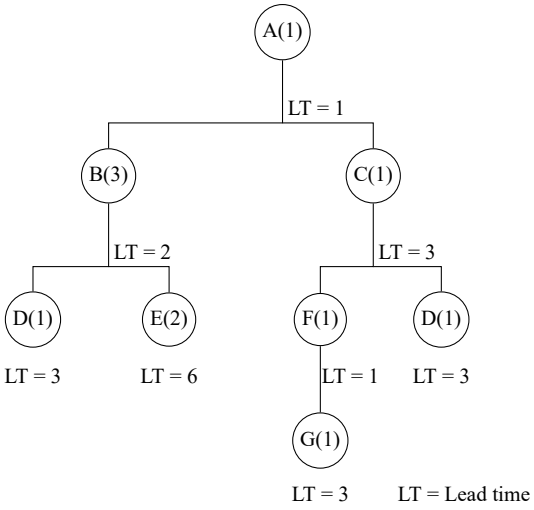
Date	**07 January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM352IA**	**Duration**	**120 Min**
Sem	**V**	**CIE – II**	
OPERATIONS MANAGEMENT			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	In services, the process structure is primarily influenced by the level of _____ and customization.	1	1	2
2	In manufacturing, a _____ process is ideal for high-volume, standardized production.	1	2	2
3	The first step in a systematic approach to long-term capacity decisions is _____ future demand.	1	2	2
4	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____.	1	1	2
5	The _____ component of a forecast captures recurring patterns over specific periods, such as seasons or months.	1	2	3
6	The smoothing constant in exponential smoothing ranges between 0 and _____.	1	1	3
7	____ provides a detailed plan of what needs to be produced and when, serving as an input for MRP.	1	1	3
8	The ____ file contains information about the quantities of materials required to make a product.	1	2	3
9	_____ integrates all business processes, including finance, HR, and supply chain, into a unified system.	1	2	3
10	_____ ensures that operational capabilities are in line with the company's strategic goals.	1	2	3
	Part –B			
1	Explain the major process strategy decisions and their implications for operations.	10	2	2

2	Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices. **a.** If other factors are equal, which two offices should be located closest together? **b.** Figure shows an alternative layout for the department. What is the total weighted-distance score for this plan based on rectilinear distance and assuming that offices A and B are three units of distance apart? **CLOSENESS MATRIX** Trips Between Departments Department: A — B 25 — C 90 — D — E 105 — F 185 B — — — — E 125 — F 125 C — — — — D — — E 25 — F — D — — — — E — — F 105 E — — — — F — F — **C F A** **B E D**	10	3	2
3	The monthly demand for units manufactured by the Acme Rocket Company has been as follows: May 100, June 80, July 110, August 115, September 105, October 110, November 125, December 120 **i.** Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; $\alpha = 0.2$. **ii.** Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December. **iii.** Calculate the tracking signal as of the end of December. What can you say about the performance of your forecasting method?	10	3	3

4	Refer to the bill of materials for product A. If there is no existing inventory and no scheduled receipts, how many units of items G, E, and D must be purchased to produce five units of end item A?  	10	4	3
5	Explain how the concept of dependent demand in material requirements planning is fundamental to resource planning.	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	–	4	6	–	4	6	–	–	–	–
	Test		–	20	30	–	–	20	20	10	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	**CIE - II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: FINANCIAL ACCOUNTING & COSTING		**Course Code : IM254TA**

Note: Answer all the Questions.

Sl. No	Questions	M	BT	CO
	PART - A			
1.1	What is meant by Overheads in Costing?	2	1	2
1.2	Differentiate between Fixed Costs and Variable Costs.	2	3	2
1.3	Identify two sectors for which activity-based costing is suitable.	2	3	4
1.4	Comment on how cost accumulation happens in Process Costing with an example.	2	3	4
1.5	Job costing is for _____, _____ jobs, while batch costing is for *a group of* _____ *products* treated as one batch.	2	2	3
	PART - B			
2.a	Enumerate the objectives of Cost Accounting.	05	2	3
2.b	Discuss the various elements of Costing.	05	2	3
3	Prepare Trial Balance for the following Balances as on 31st March 2023	10	3	2

Balances	**₹**
Cost of Goods Sold	5,20,000
Opening Stock	50,000
Closing Stock	50,000
Salary and Wages	50,000
Sales	8,00,000
Plant & Machinery	2,00,000
Drawing	50,000
Investment	4,30,000
Creditors	1,00,000
Capital	4,00,000

4.	Prepare Profit and Loss account from the following particulars for the year ended March 31, 2023: Gross Profit ₹ 4,69,000 Subletting ₹ 10,000 Commission Received ₹ 40,000 Rent, tax, and Insurance ₹ 10,000 Discount Received ₹ 22,000 Export Duty ₹ 15,000 Rent Received ₹ 23,000 Audit Fees ₹ 5,000 Bank Charges₹ 8,000 Packing Charges(Show Room)₹ 12,000 Dividend Received ₹ 15,000 Depreciation ₹ 24,000 Administration Expenses ₹ 35,000 Selling Expenses ₹ 25,000 Printing and Stationery ₹ 15,000 Distribution Expenses ₹ 28,000 Office Expenses ₹ 32,000 Trade Expenses ₹ 50,000 General Expenses ₹ 25,000 Carriage outward ₹ 15,000 Legal Expenses ₹ 5,000 Bad Debts ₹ 10,000	10	4	3
5	Write short notes on any three types of costing methods used in organizations.	10	2	3
6	In a manufacturing unit, raw material passes through four processes, I, II, III, and IV and the output of each process is the input for the subsequent process. The losses in the four processes are respectively 25%, 20%, 20% and 16 2/3 % respectively for I, II, III and IV processes of the input. If the end product at the end of the IV process is 40,000 kg, what is the quantity of raw material required to be fed at the beginning of Process I and the cost of the same at Rs. 5 per kg?	10	3	3

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**8th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM254TA**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – II**	
FINANCIAL ACCOUNTING AND COSTING			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	What is a Trial Balance, and why is it prepared?	02	2	CO2
2	How is Gross Profit calculated?	02	1	CO3
3	Why is depreciation included in the Income Statement?	02	2	CO3
4	What is the basic accounting equation represented in a Balance Sheet?	02	1	CO3
5	What is the difference between current and non-current assets?	02	1	CO1

PART – B

Sl. No	Questions	M	BT	CO
1.	The following are the ledger balances of a business as of December 31, 2024. Prepare a trail balance.	10	3	CO2

Account	Amount (₹)	Type
Cash	50,000	Debit
Capital	1,50,000	Credit
Sales	2,00,000	Credit
Purchases	1,00,000	Debit
Furniture	25,000	Debit
Salaries	30,000	Debit
Accounts Receivable	45,000	Debit
Accounts Payable	40,000	Credit

Academic year 2024-2025 (ODD Sem)

2.	Prepare a Profit and Loss Account for Green Enterprises for the year ending December 31, 2024, from the following details: **Particulars** / **Amount (₹)** Sales Revenue 7,50,000 Purchases 4,00,000 Opening Inventory 1,20,000 Closing Inventory 80,000 Wages 50,000 Office Expenses 40,000 Utilities 30,000 Depreciation on Machinery 20,000	10	4	CO3
3.	Prepare a Balance Sheet for Green Valley Enterprises as of December 31, 2024, from the following details: **Particulars** / **Amount (₹)** / **Particulars** / **Amount (₹)** Capital 5,00,000 / Cash 1,00,000 Bank Loan 3,00,000 / Building 5,00,000 Accounts Payable 1,50,000 / Inventory 2,00,000 Salaries Payable 50,000 / Furniture 80,000 Accounts Receivable 1,20,000 / Equipment 1,50,000	10	3	CO3
4.	From the following information, prepare a cost sheet for XYZ Manufacturing Co. for the year ending March 31, 2025: **Particulars** / **Amount (₹)** Raw Materials Consumed 3,00,000 Direct Labor 2,00,000 Factory Overheads 80,000 Office and Administrative Overheads 40,000 Selling and Distribution Overheads 30,000 Opening Stock of Finished Goods 50,000 Closing Stock of Finished Goods 60,000 Profit Percentage on Cost 25%	10	3	CO4
5.	Explain the three main elements of cost sheet with examples. How are they classified and used in cost accounting?	10	3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	02	02	06	–	–	06	04	–	–	–	–
		Test	–	10	20	20	–	–	–	30	20	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	**CIE-II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Quality Assurance		**Course Code : IM353IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Explain the purpose of a p-chart.	2	3	01
2.	What is AOQ?	2	2	01
3.	State the difference between single and double sampling plans.	2	2	02
4.	What is AOQL?	2	3	02
5.	What are multiple sampling plans?	2	2	02
	6.			
7.	Draw the type - A OC curve for the single sampling plan: $n = 50$, $C = 1$, $N = 200$.	10	2	3
2	Differentiate between C chart and U chart	10	3	3
3	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results. Lot No. / Amount in lot / Number rejected 01 / 6,000 / 183 02 / 6,000 / 131 03 / 10,000 / 184 04 / 1,783 / 23 05 / 3,089 / 171 06 / 5,774 / 65 07 / 3,000 / 46 08 / 2,724 / 85 Draw an appropriate control chart for the data. What can you conclude on the state of control?	10	4	3

<table>
<tr><td>4</td><td>The following table gives the number of missing rivets at the aircraft final inspection:
<table>
<tr><th>Sl No</th><th>Air craft No.</th><th>Number of missing rivets</th><th>Sl No</th><th>Air craft No.</th><th>Number of missing rivets</th></tr>
<tr><td>1</td><td>202</td><td>07</td><td>6</td><td>207</td><td>11</td></tr>
<tr><td>2</td><td>203</td><td>12</td><td>7</td><td>208</td><td>06</td></tr>
<tr><td>3</td><td>204</td><td>13</td><td>8</td><td>209</td><td>19</td></tr>
<tr><td>4</td><td>205</td><td>18</td><td>9</td><td>210</td><td>04</td></tr>
<tr><td>5</td><td>206</td><td>17</td><td>10</td><td>211</td><td>26</td></tr>
</table>
Establish an appropriate control chart for the purpose.</td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>5</td><td>Explain the meaning of the following terms:
<table>
<tr><td>i.</td><td>Producer’s risk</td><td>ii.</td><td>AQL</td></tr>
<tr><td>iii.</td><td>Consumer’s risk</td><td>iv.</td><td>LTPD</td></tr>
</table></td><td>10</td><td>3</td><td>4</td></tr>
</table>

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**7th January 2025**	**Maximum Marks**	**50**
Course Code	**IM353IA**	**Duration**	**90 Min**
Sem	**V**	**CIE – II**	
QUALITY ASSURANCE			

Note:
1. Answer all the Questions.

PART – A **Max Marks: 10**

Sl. No	Questions	M	BT	CO
1.	Differentiate between Attribute control chart and Variable control chart.	2	L2	CO1
2.	What does an R-chart measure, and why is it important in statistical process control?	2	L1	CO2
3.	What is the key difference between a p-chart and a np-chart?	2	L2	CO3
4.	Expand the acronyms: AQL, UTPD	2	L2	CO3
5.	Write two applications of acceptance sampling.	2	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO
1.	A TiW layer is deposited on a substrate using a sputtering tool. The following table contains layer thickness measurements (in Angstroms) on 10 subgroups of four substrates. Set up $\bar{X}$ and s charts on this process and compute trial control limits.	10	3	CO3
2.	Write a note on Standardized control chart. What is the process capability? Explain any one of the techniques of process capability analysis.	10	2	CO1

Subgroup	x_1	x_2	x_3	x_4
1	459	449	435	450
2	443	440	442	442
3	457	444	449	444
4	469	463	453	438
5	443	457	445	454
6	444	456	456	457
7	445	449	450	445
8	446	455	449	452
9	444	452	457	440
10	432	463	463	443

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>3.</td><td>Lots of Bushing screws from a single supplier were inspected 100 percent with the following results.
<table>
<tr><th>Lot No.</th><th>Amount in lot</th><th>Number rejected</th></tr>
<tr><td>1</td><td>6000</td><td>183</td></tr>
<tr><td>2</td><td>6000</td><td>131</td></tr>
<tr><td>3</td><td>10000</td><td>184</td></tr>
<tr><td>4</td><td>1783</td><td>23</td></tr>
<tr><td>5</td><td>3089</td><td>171</td></tr>
<tr><td>6</td><td>5774</td><td>65</td></tr>
<tr><td>7</td><td>3000</td><td>46</td></tr>
<tr><td>8</td><td>2724</td><td>85</td></tr>
</table>
Draw an appropriate control chart for the data. What can you conclude on the state of control?</td><td>10</td><td>2</td><td>CO2</td></tr>
<tr><td>4.</td><td>Draw the OC Curve, AOQ Curve, ASN Curve and ATI Curve for the single sampling plan with N = 5000, n = 100 and c = 2</td><td>10</td><td>4</td><td>CO4</td></tr>
<tr><td>5.</td><td>A factory inspects different samples of a product, recording the number of defects in each sample. The sample size is constant, and the number of defects is provided below:
<table>
<tr><th>Sample No.</th><th>Number of Defects</th></tr>
<tr><td>1</td><td>5</td></tr>
<tr><td>2</td><td>8</td></tr>
<tr><td>3</td><td>6</td></tr>
<tr><td>4</td><td>4</td></tr>
<tr><td>5</td><td>7</td></tr>
<tr><td>6</td><td>9</td></tr>
<tr><td>7</td><td>5</td></tr>
<tr><td>8</td><td>6</td></tr>
</table>
i. Draw an appropriate control chart for the data. Evaluate the process.
ii. Plot the centreline, upper control limit (UCL), and lower control limit (LCL).</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Quiz	Max Marks	02	04	04	–	2	8	–	–
	Test		10	20	10	10	–	30	10	10

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 17/12/2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define Robotic Process Automation (RPA) and state its primary focus.	2	1	CO1
2.	State the fundamental reason why a company must engage in Business Process Reengineering (BPR) before implementing any major automation initiative.	2	2	CO2
3.	Define the term Data Mart and state how it differs in **scope** from a full Data Warehouse.	2	1	CO3
4.	Name the two foundational processes that are typically executed by Data Mining tools to discover new knowledge within a BI system.	2	2	CO3
5.	A decision-maker uses a BI tool to analyze sales by clicking on a region, then a city, and finally a specific store. Name this OLAP technique and briefly state its purpose.	2	2	CO3
	Part – B			
1	List and explain various methods of business process automation.	10	2	CO2
2	Define Robotic Process Automation (RPA). Explain how it works. List the uses and benefits.	10	2	CO2
3	Explain the core difference between Business Intelligence (BI) and Data Warehousing. Describe why BI is considered “the technology and practice of applying information to make decisions”.	10	2	CO1
4	Describe the four progressive types of Data Analytics used in an EIS. For each type, state the central question it is designed to answer.	10	3	CO3
5	Apply the concept of Online Analytical Processing (OLAP) by explaining its primary function. Illustrate how a retail manager would use an OLAP technique like Drilling Down to analyze a regional sales report.	10	4	CO3

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**9th January 2025**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**V**	**CIE – 2**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	Mention the situations that make the replacement of items necessary?	02	2	1
2.	What are the three strategies of replacement of items which follow sudden failure mechanisms?	02	2	3
3.	State two differences between preventive maintenance and breakdown maintenance.	02	2	2
4.	Give one advantage and one disadvantage of individual replacement policies.	02	2	2
5.	State the main objectives of using the EOQ model?	02	1	3
	Part B			
1	A firm is considering the replacement of a machine, whose cost price is Rs 12,200, and its scrap value is Rs 200. From experience the running (maintenance and operating) costs are found to be as follows: Year: 1 2 3 4 5 6 7 8 Running Cost (Rs): 200 500 800 1200 1800 2500 3200 4000 When should the machine be replaced?	10	3	1
2	The data on the running costs per year and resale price of equipment A, whose purchase price is Rs 2,00,000 are as follows: Year: 1 2 3 4 5 6 7 Running cost (Rs): 30,000 38,000 46,000 58,000 72,000 90,000 1,10,000 Resale value (Rs): 1,00,000 50,000 25,000 12,000 8,000 8,000 8,000 i. What is the optimum period of replacement? ii. When equipment A is two years old, equipment B, which is a new model for the same usage, is available. The optimum period for replacement is 4 years with an average cost of Rs 72,000. Should equipment A be changed with equipment B? If so, when?	10	3	2

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>3</td><td>A company is considering the purchase of a new machine at Rs 15,000. The economic life of the machine is expected to be 8 years. The salvage value of the machine at the end of the life will be Rs 3,000. The annual running cost is estimated to be Rs 7,000.
i. Assuming an interest rate of 5 per cent, determine the present worth of future costs of the proposed machine.
ii. Compare the new machine with the presently-owned machine that has an annual operating cost of Rs 5,000 and cost of repair Rs 1,500 in the second year, with an annual increase of Rs 500 in the subsequent years of its life.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>4</td><td>The following failure rates have been observed for a certain type of bulbs.
<table>
<tr><td>End of Week:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Probability of failure to date:</td><td>0.05</td><td>0.13</td><td>0.25</td><td>0.43</td><td>0.68</td><td>0.88</td><td>0.96</td><td>1.0</td></tr>
</table>
The cost of replacing an individual bulb is Rs. 2.25. if the cost of group replacement is 60 paisa per bulb and the total number of bulbs is 1000. What is best interval between group replacements?</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>5</td><td>Differentiate between Continuous Review and Periodic Review Inventory Systems by the following aspects:
i. Monitoring
ii. Reorder Trigger
iii. Stockouts Risks
iv. System Complexity
v. Suitability</td><td>10</td><td>2</td><td>2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	1	2	2	–	1	2	2	–	–	–
	Test	Max Marks	10	20	20	–	–	10	40	–	–	–

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RV COLLEGE OF ENGINEERING®, Bengaluru-59

(Autonomous Institution Affiliated to VTU, Belagavi)

Program	CHEMICAL ENGINEERING			
Course Title	GREEN AND HYDROGEN TECHNOLOGY			
Course Code	CH375TGD			
CONTINUOUS INTERNAL EVALUATION - 2				
GLOBAL ELECTIVE: GROUP D				
Sem	V	**Date | Time**	**DEC 2025**	
Max Marks	**60**	**16.12.2025 | 13:20 - 15:20**	**Max Duration**	**120 Min**

Instructions to Candidates:

All the questions are compulsory. No additional booklet will be provided.

Q. No.	QUESTIONS	M	BT	CO
1.1	What are the outcomes of hydrogen risk assessment?	02	01	01
1.2	List the steps involved in the quantitative hydrogen risk assessment.	02	01	04
1.3	Define Codes of Practice.	02	01	03
1.4	What is PAS?	02	01	02
1.5	List the parameters associated with hydrogen for which the codes and standards are defined. Hint: The approval for these parameters vary between countries.	02	01	02
Part B				
2	Explain in detail, how hydrogen gas is stored as compressed gas including its advantages and disadvantages.	10	02	04
3	What are the salient features of storing hydrogen as cryogenic liquid? How is it done and what are the associated benefits and hurdles?	10	02	04
4	Is storing hydrogen as a metal hydride a boon or a bane (annoyance)? Justify with substantial details including the working principle, advantages and disadvantages.	10	03	03
5	How are liquid organic hydrogen carriers beneficial in case of storage and transport of hydrogen? Provide a detailed narration.	10	02	03
6	Give an explanatory note on the codes and standards (national and international) associated with the handling and safety of hydrogen gas.	10	03	01

Course Outcomes

At the end of the course the student will be able to,	
1	Understand the importance of hydrogen and its use as an energy carrier
2	Explain the production, storage and handling of hydrogen
3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
4	Appraise the importance of safety, regulations and codes

Cos/BTL	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks	12	04	22	22	10	30	20	.	.	.